



Study shows how to reduce physical risk in VR

A recent study from Oregon State University has assessed how some common virtual reality (VR) movements contribute to muscle strain and discomfort. It is an effort to ensure future user safety in this fast-growing technology that is used not only for gaming but also increasingly for education and industrial training.

Researcher Jay Kim of OSU'S College of Public Health and Human Sciences said, "There are no standards and guidelines for virtual and augmented reality (AR) interactions. We wanted to evaluate the effects of target distances, locations and sizes so we can better design these interfaces to reduce the risk for potential musculoskeletal injuries."

VR users wear a headset and engage in full-body, 3D movements—unlike conventional computer users, where a desk or arms of a chair offer some level of support for hands and arms.

With sensors placed on participants' joints and muscles, researchers used motion capture to record their movements and electromyography to measure electrical activity in their muscles while performing common VR gestures. Wearing an Oculus Rift VR headset, participants were tasked with either pointing to specific dots around a circle, or colouring in a certain area with their finger.

Researchers repeated the tests with visuals placed at eye level, fifteen degrees above eye level, fifteen degrees below eye level and thirty degrees below eye level. Regardless of the angle, extending the arm straight out caused shoulder discomfort in as little as three minutes. With prolonged use, as VR often requires, this may lead to major health problems like gorilla arm syndrome and rotator cuff injuries. In addition, the heavy VR headset may increase the burden on the cervical spine, risking greater neck strain.

Researchers focused on neck and shoulder movements. They found performance in the colouring task was worst when participants had to tilt their heads down fifteen and thirty degrees. The most extreme postures and highest muscle activity were observed with targets at fifteen degrees above eye level, as participants were forced to constantly maintain their extended neck and elevated arm position. And discomfort was greatest in the pointing task at fifteen degrees above eye level.

Kim said, "Based on this study, we recommend that objects that are being interacted with more often should be closer to the body. And objects should be located at eye level, rather than up and down."

VR, drones help to predict and protect koala habitat

Researchers from Queensland University of Technology and ARC Centre of Excellence for Mathematical and Statistical Frontiers (ACEMS) have used the mashup of high-tech 360-degree imagery and heat-seeking drone cameras along with traditional techniques of ground surveys to develop a model that could be used to identify areas most likely to be home to koalas, which are facing population decline, especially after the recent fires that ravaged their habitat in Australia.

Lead author Dr Catherine Leigh, said, it is about building a model that tells us with confidence where koalas are and where they are not. When you start building up a model with all of that data, you get a much better idea about where the koalas are likely to be. And if you can predict where koalas are, and the types of habitats they are hanging around in, then you know not only where to go to keep monitoring them but also the areas you need to protect.

In the study, researchers used ground survey information captured between 2012 and 2017 from Alexander Clarke Park in Loganholme, south-east Queensland, which is an area that has vegetation ranging from dense forest to grass fields, and contains facilities including playgrounds and a dog park.

Dr Leigh said one of the challenges with ground survey observations was that they could show a statistical bias of higher koala populations close to paths and other areas easily accessible by observers.

She says, "When you are out in the field as an experienced koala observer or a citizen scientist, you really are restricted where you can walk. And koalas are hard to see, even by trained observers. They are slow moving and often hidden high in the canopy."

To counteract that, researchers surveyed the area from the air using thermal cameras near sunrise, when body heat of the koalas is detectable against the surrounding foliage.

The next part of the study involved sending researchers out to some 82 specific locations in the area where koalas were known to either be or not be, to take 360-degree images. Those images were then shown to a panel of koala experts wearing virtual reality (VR) headsets, with experts asked to state how likely the environment they were viewing in was good koala habitat.

Dr Leigh said the advantage of using VR for the survey was that it enabled them to bring the environment to the experts rather than having to take the experts to the field.